

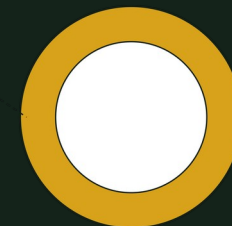


DEALER TRAINING · SEPTEMBER 2026

VL-Potash Liquid

Liquid potash, K_2O 30 g in every 100 ml, on the leaf or down the line at the stage the crop fills.

For VAN dealers and the field team, written so a dealer can repeat it to a grower. Lahore.



What a dealer says in 30 seconds.

VL-Potash Liquid is potash in a bottle: K_2O 30 g in every 100 ml, 300 g in a litre, made so it stays in solution in alkaline water. The crop grows on nitrogen and fills on potash, and only 1% of the nutrient Pakistan applies is potash. A liquid can be put on at the moment the crop asks, on the leaf or through the drip line, instead of months before it. The published programmes spray 1 bottle an acre on wheat at early growth and again at grand growth, 3 bottles on sugarcane at maturity, 1 bottle on tomato after each picking, 1 on guava at flowering and again in December, and run 1 through the water on mango at grand growth. It is a correction at the fill window, in grams. The season's potash is the basal bag in the ground.

The 4 things it must contain

1

The bottle: 1 L, K_2O 30% w/v, PS 5469-2020. 300 g of K_2O in every litre. 3 L, 20 L and 200 L packs too.

2

The nutrient: potash moves the sugar into the grain, cane, tuber or fruit. 1% of Pakistan's applied nutrient is potash.

3

The stage: when the crop fills. On the leaf or through the line, at the row your crop's plan prints.

4

The limit: grams at the peak, not the season's potash. The basal bag went in at land preparation.

Say it in the grower's own words. Every figure in these 30 seconds is on the bag, in the published crop programme, or in the named evidence on the slides that follow.

THE BAG

What the bottle says, and what it is registered as.

1 L

the bottle. 300 g of K₂O in every litre, from the declared analysis. 3 L, 20 L and 200 L packs are on the price sheet.

K₂O 30%

w/v: 30 g in every 100 ml. The programme rows print it as "Potash 30%".

pH-regulated

phosphorus and boron are added to hold the pH of the solution, as VAN's product data says. Named for their job, not as headline nutrients, and never numbered.

2 routes

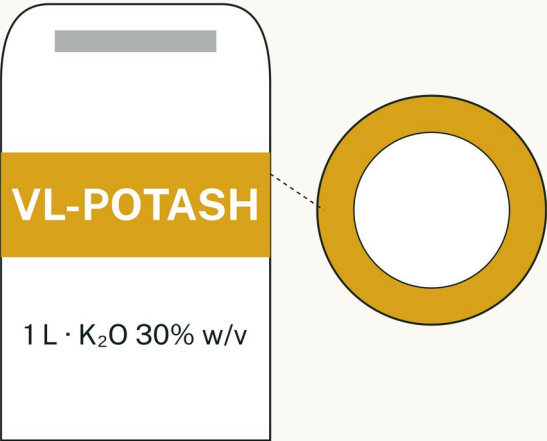
foliar and fertigation, as VAN names them. The programme row says which for your crop.

1%

of all nutrient applied in Pakistan is potash, as the Crop Force bag prints. About 1 kg for every 83 kg of nitrogen.

1 helpline

the drone grade of VAN's potash is a separate product with its own deck; no drone rate is in this one.



Registered as	Pakistan Standard	Manufacturing licence
VL-Potash Liquid	PS 5469-2020	CM/L-2943/2023

No expiry date is printed in this deck. The licence number is what a dealer quotes.

A GROWER'S QUESTION

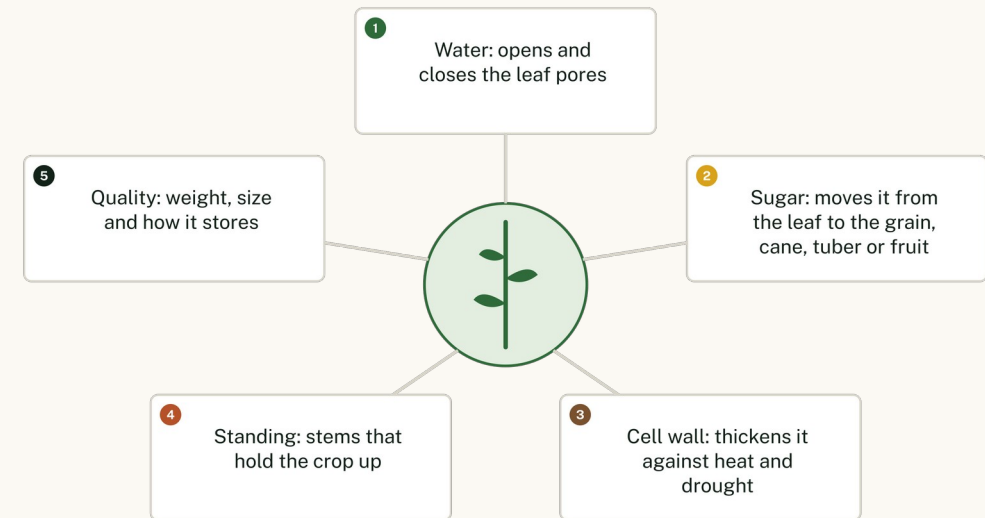
“I put the potash in at sowing. When the crop started to fill it looked short again. Can I give potash to a standing crop?”

The grower did the basal potash. What he is asking is whether potash can arrive at the moment the crop asks for it, weeks after the bag went in. The answer is 3 ideas: what potash does at fill, why a liquid can be there at the peak, and how it reaches the crop past the soil.

The 3 answers on the next 3 slides, 1 idea each.

The crop grows on nitrogen and fills on potash.

Potash runs the plant's plumbing. It opens and closes the pores in the leaf, moves the sugar the leaf makes into the grain, the cane, the tuber or the fruit, and thickens the cell wall against heat and drought. A crop short of it shows scorched edges on the older leaves, weak stems, light grain and small fruit that stores badly. Only 1% of all nutrient applied in Pakistan is potash, about 1 kg for every 83 kg of nitrogen, so on most fields the soil's reserve is doing the filling alone.

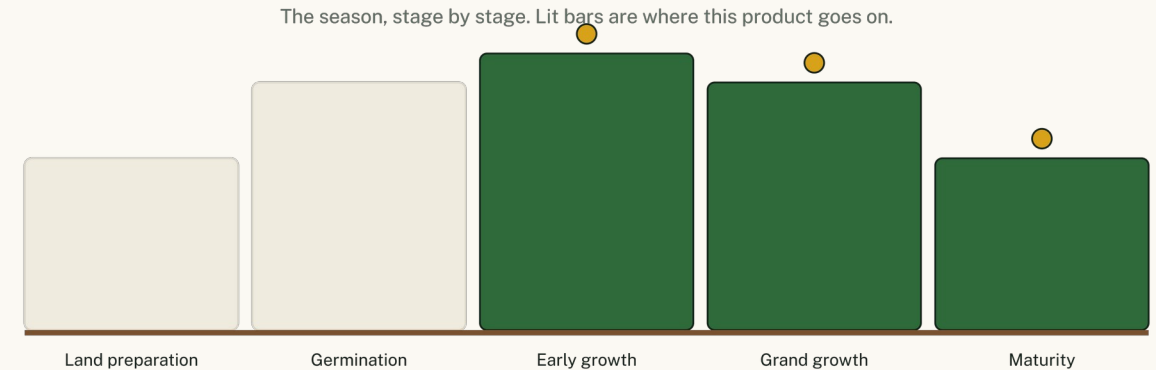


The 5 jobs of potash, as VAN's 17-nutrients page gives them. A picture of the idea, not a measurement.

Say it like this: "You fed the growing. This is for the filling."

A liquid can be there at the peak, not months before it.

A basal bag goes in at land preparation and the crop draws on it for the season. Potash demand peaks when the crop fills, and a liquid can be put on in that window and repeated when the crop asks again. The programmes do exactly that: wheat at early growth and grand growth, sugarcane at maturity, tomato after each picking, guava at flowering and again in December, mango at grand growth. It covers the demand as it happens. The basal bag covers the base.

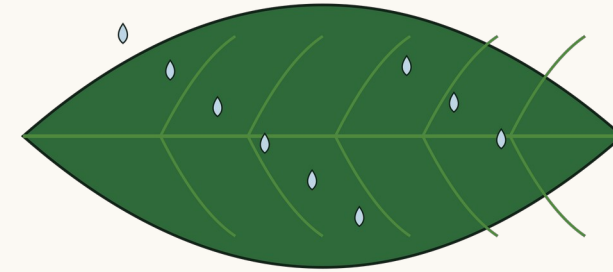


The 5 stage names are the programme's. Which rows carry VL-Potash Liquid for your crop are on the programme slides.

Say it like this: "The bag in the ground is the base. The bottle is for the weeks the crop is filling."

On the leaf or through the line: past the soil.

On calcareous ground above pH 8, what a granule releases meets the soil chemistry first. A spray puts potash on the leaf, where it is taken up directly while the droplet is wet; a fertigation run carries it in the water to the wet root zone. Alkaline irrigation water would drop an unregulated potash solution out of solution, so VAN's formulation carries phosphorus and boron to hold the pH. Both routes are fully soluble and leave no residue in a drip emitter, as VAN's product data says.



Sprayed onto the leaf. Small doses, taken up through the leaf surface, for a nutrient the crop needs quickly or that the soil locks up. Spray early morning or late afternoon; never on a wilted crop.

Say it like this: "It goes on the leaf or down the line. It does not sit in the soil waiting to be locked."

The leaf taking up the droplet. The fertigation route is drawn on the method slide.

What is verified for VL-Potash Liquid, and what is not.

No trial of VL-Potash Liquid is on VAN's record. What a dealer may state as fact is the bottle, the licence, the programme rows, the national figure, the soil record and the published potash research below, each as what it is.

Item	What it says	How a dealer may use it
The bottle	VL-Potash Liquid, K ₂ O 30% w/v (300 g/L), with phosphorus and boron added for pH regulation of the solution; 1 L, 3 L, 20 L and 200 L on the price sheet.	As fact. P and B named for their job, never numbered.
The licence	Pakistan Standard PS 5469-2020 for liquid nutrient supplements; manufacturing licence CM/L-2943/2023.	As fact. The number, never an expiry date.
The programme rows	9 rows on 6 programmes: 1 pack an acre, spray, on wheat twice and guava twice (guava at 1 L per 100 L of water); 3 packs on sugarcane and ratoon at maturity; 1 pack after each picking on tomato; 1 pack by fertigation on mango.	As the published rate, as printed.
National potash use	Only 1% of all nutrient applied in Pakistan is potash (the Crop Force bag). About 1 kg for every 83 kg of nitrogen (VAN's national figure).	As fact.
The soil record (Punjab, 770,160 samples)	87.8% of samples below the working threshold for available potash; mean 132 ppm. 69.1% above pH 8.	As fact about the soil. Never as product performance.
Published research, not VAN trials	Sindh banana (Memon et al. 2010): potash applied far below removal; correcting potash alone lifted yield from 51.2 to 60.8 t/ha. Multan cotton (Hussain, Ali and Gardezi 2021): soil potassium reduced boll shedding.	Spoken, as research on potash in the soil. Never as a VL-Potash Liquid result.
Not on file	A VL-Potash Liquid trial; a yield, sugar-recovery or quality figure; a drone rate; a Pakistani uptake curve for potato or banana.	Not said until named.

Sources: the bottle; the Pakistan Standard and licence register; the crop programmes on van.com.pk; the VAN farmer price sheet, Aug 2026; the Punjab soil record (770,160 samples); Memon et al. 2010; Hussain, Ali and Gardezi 2021.

The 4 packs, in grams of K₂O, in the grower’s own back-of-envelope.

Prices from the VAN farmer price sheet, Aug 2026, delivered farmer price. K₂O from the declared analysis, w/v: 30% is 300 g in a litre.

VL-Potash Liquid, K ₂ O 30% w/v	Volume	K ₂ O	Rupees	Rupees per litre
1 bottle	1 L	300 g	1,250	1,250
1 pack	3 L	900 g	3,000	1,000
1 pack	20 L	6.0 kg	18,500	925
1 drum	200 L	60 kg	175,000	875

A bottle is 300 g of K₂O on the leaf or in the line, for 1,250 rupees. The 3 L pack is the sugarcane row in 1 pack. The 20 L and 200 L packs are for a dealer or a large farm; the grower buys the pack his plan’s bottle count fills.

VL-Potash Liquid 1 L PKR 1,250, 3 L PKR 3,000, 20 L PKR 18,500, 200 L PKR 175,000: the VAN farmer price sheet, Aug 2026. 300 g = 1,000 ml × 0.30; 6.0 kg = 20 × 0.30. Rupees per litre = price ÷ litres.

What the programme puts on, per acre, per season.

Bottles per acre from the published programmes; K₂O from the declared analysis; rupees at the VAN farmer price sheet, Aug 2026, 1 L bottles.

Crop	Stage and method, as printed	Bottles	K ₂ O	Rupees per acre
Wheat	Early growth, spray; grand growth, spray	2	600 g	2,500
Sugarcane	Maturity, spray	3	900 g	3,750 (3,000 as the 3 L pack)
Sugarcane ratoon	Maturity, spray	3	900 g	3,750 (3,000 as the 3 L pack)
Tomato	Fruiting and picking phase, spray, 1 pack after each picking	1 per picking	300 g per picking	1,250 per picking
Mango	Grand growth, fertigation	1	300 g	1,250
Guava	October (flowering starts) and December, foliar, 1 L per 100 L water	2	600 g	2,500

On wheat the grower spends 2,500 rupees an acre to put 600 g of K₂O on the leaf across the 2 stages the plan names. The guava row is the 1 programme that prints the water volume: 1 L per 100 L. Tomato is counted by pickings, so the season's total is the grower's picking count.

Rows from the crop programmes on van.com.pk, as printed. VAN's wheat cost plan carries the same 2 L an acre at 1,250. Sugarcane $3 \times 1,250 = 3,750$; the 3 L pack is 3,000 on the same sheet.

The wheat potash account: what leaves, what the 2 bottles put on.

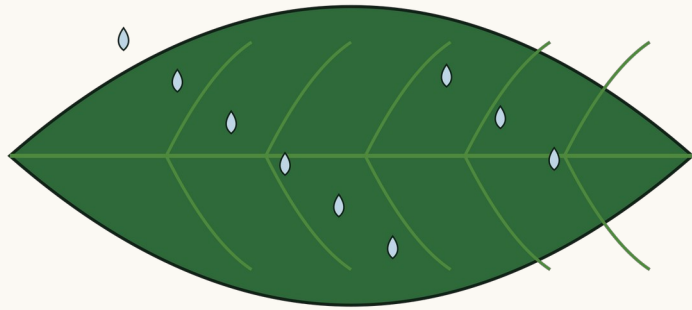
What 40 maunds of wheat (1.6 tonnes) takes off the field, from the IPNI removal tables VAN’s balance block uses, beside the 2 sprays the programme puts on.

Wheat, 1 acre at 40 maunds	K ₂ O per tonne of grain	K ₂ O at 1.6 t	
Grain taken off	4.8 kg	7.7 kg	
Straw (bhusa) taken off too, per tonne of grain	20 kg	32 kg	
Grain and straw together	24.8 kg	40 kg	
VL-Potash Liquid, 2 bottles, early and grand growth, puts on		0.6 kg	

The 2 bottles put 600 g of K₂O on the leaf at the 2 stages that build tillers and fill grain. The crop takes about 40 kg off the field with the bhusa. The bottle is a correction at the moment of demand; the season’s potash is the basal bag and the soil’s reserve, and 87.8% of Punjab samples already read low.

HOW IT GOES ON

On the leaf, or through the line. Water first, then the bottle. Morning or evening.



Sprayed onto the leaf. Small doses, taken up through the leaf surface, for a nutrient the crop needs quickly or that the soil locks up. Spray early morning or late afternoon; never on a wilted crop.

Spray: water first, product second

Fill the tank with clean water, add the bottle, keep it agitated. The guava row prints the rate a plan names: 1 L per 100 L of water. VAN's general guidance is 100 to 200 L of water an acre; the dilution is part of the rate, and cutting the water doubles what lands on the leaf. A fine, even fan, to cover without run-off. Morning or evening, never midday.

Fertigation: dissolve, then release with the water flowing

The mango row runs 1 bottle through the water at grand growth. Into the drum or the drip tank, stirred in, released across the whole irrigation and not in 1 slug. Fully soluble, no residue in the emitter, as VAN's product data says.

At the row your plan prints

Wheat at early growth and grand growth. Sugarcane and ratoon, 3 bottles at maturity. Tomato, 1 bottle after each picking. Guava at flowering and in December. Never broadcast, never in place of the basal bag, and never at a drone rate from this deck.

“1,250 for 1 litre of potash? A bag of SOP is 8,000 for 25 kg.”

The dealer says, out loud

They are 2 different jobs. The bag goes in the ground at land preparation and the crop draws on it all season; the bottle goes on the leaf or down the line at the moment the crop is filling. On wheat it is 2 bottles, 2,500 rupees, at early growth and grand growth: 600 g of potash arriving exactly when the tillers and the grain want it. Write both on your paper as 2 lines, not 1 choice. The plan uses both.

The fact behind it

- VL-Potash Liquid 1 L 1,250 rupees; 3 L 3,000; 20 L 18,500; 200 L 175,000 (VAN farmer price sheet, Aug 2026). SOP 25 kg 8,000 on the same sheet.
- 300 g K₂O per litre, from the declared analysis (the bottle).
- Wheat: 2 bottles an acre, early growth and grand growth, spray (the crop programmes on van.com.pk).
- The basal potash is a different bag on the same programme.

Never say: “It gives you X more maunds” or a rupees-per-acre return. No yield figure exists for the product. No cost per kg of K₂O against the bag.

“Can I skip the basal potash and spray this instead?”

The dealer says, out loud

No. A litre is 300 g of potash. The crop takes tens of kilograms off an acre; on wheat about 40 kg with the bhusa. The bottle is a correction at the peak, in grams; the basal bag is the season's potash, in kilograms. Keep the bag in the ground at land preparation and add the bottles at the stages the plan names.

The fact behind it

- Wheat at 40 maunds takes off 7.7 kg K₂O in the grain and about 40 kg with the straw (the IPNI removal tables).
- 2 wheat bottles = 600 g K₂O.
- Every VL-Potash Liquid row sits beside a basal potash row on the same programme.

Never say: “2 sprays and you can skip the bag.”

“Will it burn the leaf?”

The dealer says, out loud

Not at the rate and the water the plan gives. The guava row says 1 L per 100 L of water; VAN's general guidance is 100 to 200 L an acre. Burn happens when the water is cut to save a pass and when the spray goes on at midday. Water first, product second, keep it agitated, a fine fan to cover without run-off, morning or evening.

The fact behind it

- Guava row: "Foliar, 1 L per 100 L water" (the crop programmes on van.com.pk).
- 100 to 200 L of water an acre, morning or evening, never midday (VAN guidance).
- Cutting the water in half doubles the concentration on the leaf.

Never say: A lower water volume, or “it cannot burn”.

OBJECTION 4 OF 8

“Can I put it in my drone?”

The dealer says, out loud

Not from this deck. VAN makes a low-volume drone grade of its potash, built for a 20 L tank, and it has its own name, its own specification and its own deck. An ordinary foliar grade at drone concentration is not the same product. Ask VAN for the drone product and its rate; I will not read one off this bottle.

The fact behind it

- VAN’s drone grade of potash is a separate product with its own specification and deck.
- VAN’s ultra-low-volume convention is 1 L of product per 20 L tank per acre, for the drone products only.
- An ordinary foliar grade at drone concentration is not the same product.

Never say: “Use the 1 L bottle in the drone at 1 L an acre.” No drone rate from this deck.

“Which stage, and how many bottles?”

The dealer says, out loud

What the plan prints. Wheat: 1 bottle at early growth and 1 at grand growth, sprayed. Sugarcane and ratoon: 3 bottles at maturity, sprayed. Tomato: 1 bottle after each picking. Guava: 1 bottle at flowering in October and 1 in December, at 1 L per 100 L of water. Mango: 1 bottle through the water at grand growth. Not at sowing: that is the basal bag's stage.

The fact behind it

- Wheat 2 × 1 pack, spray; sugarcane and ratoon 3 packs at maturity, spray (the crop programmes on van.com.pk).
- Tomato 1 pack after each picking; guava 1 pack in October and December; mango 1 pack by fertigation at grand growth (the crop programmes on van.com.pk).
- VAN's product entry names maturity as the stage; the wheat rows add early and grand growth.

Never say: “More bottles, more yield”, or a rate for a crop with no row.

“Is it registered? What is the licence?”

The dealer says, out loud

Pakistan Standard PS 5469-2020 for liquid nutrient supplements, manufacturing licence CM/L-2943/2023, VAN as the registered maker. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Read the batch number off the bottle and VAN sends you that batch's certificate, free.

The fact behind it

- VL-Potash Liquid: PS 5469-2020, CM/L-2943/2023 (the licence register on van.com.pk).
- Every batch is tested at VAN Lab, PNAC LAB 336, before it ships.
- The certificate for any batch is free on request against the batch number.

Never say: An expiry date. The number is what a dealer quotes.

“Can it go through the drip on my other crops, not just mango?”

The dealer says, out loud

VAN names 2 routes for this bottle, foliar and fertigation, and the mango row is the 1 programme that prints fertigation. It is fully soluble and leaves no residue in the emitter. For a crop whose row says spray, the plan's rate is a spray rate; if you want to run it through the line, ask VAN for the fertigation rate for that crop rather than reading the spray bottle count into the tank.

The fact behind it

- VAN names foliar and fertigation as the routes for VL-Potash Liquid.
- Fertigation is printed on the mango row only (the crop programmes on van.com.pk).
- The fertigation grade is fully soluble and leaves no residue in an emitter (VAN product data).

Never say: A fertigation rate the plan does not print.

“My cane is standing tall but the mill says the recovery is low. Will this fix it?”

The dealer says, out loud

Potash moves the sugar from the leaf into the stalk; that is its job, and the sugarcane programme puts 3 bottles on at maturity for that stage. Put it on as the plan says. I will not promise you a recovery figure, because VAN has not measured one for this product, and a mill result has many causes besides potash.

The fact behind it

- Potash moves sugar from the leaf to the grain, cane, tuber or fruit (VAN's 17-nutrients page).
- Sugarcane and ratoon: 3 packs at maturity, spray (the crop programmes on van.com.pk).
- No sugarcane trial of VL-Potash Liquid is on VAN's record.

Never say: “It raises recovery by X”, “it ripens the cane”, or any sucrose figure.

VL-Potash Liquid in the published programmes, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Early Growth	1 pack	Spray	1 L
Wheat	Grand Growth	1 pack	Spray	1 L
Sugarcane, February planting	Maturity	3 pack	Spray	1 L
Ratoon Sugarcane	Maturity	3 pack	Spray	1 L
Tomato	Fruiting and Picking Phase	1 pack after each picking	Spray	1 L
Mango	Grand Growth	1 pack	Fertigation	1 L
Guava	October, flowering starts	1 pack	Foliar, 1 L per 100 L water	1 L
Guava	December	1 pack	Foliar, 1 L per 100 L water	1 L

Rows as published in van.com.pk's crop programmes, all 6 programmes that name VL-Potash Liquid; spray on 5, fertigation on mango. Guava prints the water volume. A rate is per acre; the crop page scales it to the field.

WHAT VAN WILL NOT CLAIM FOR IT

A dealer who says any of these is speaking for himself, not for VAN.

1

A yield, sugar-recovery, firmness or shelf-life figure. No VL-Potash Liquid trial exists.

2

That it replaces the basal potash or is the season's potash. It is grams at the peak; the bag in the ground is the base.

3

A drone rate, or the drone grade's name, from this deck. It is a separate product with its own deck.

4

A disease, cold or stress-tolerance claim, or a nitrogen-efficiency claim. Potash thickens the cell wall; nothing is measured for this product.

5

A phosphorus or boron figure. They are in the bottle to hold the pH, named for that job and never numbered.

6

A water volume or a fertigation rate the plan does not print, as if it were a programme figure.

Every batch is tested. The bag carries the proof.

1 · The batch number

It is printed on the bag. Read it to the grower and show him where it is. Every bag of every batch carries one.

2 · The certificate of analysis

For any batch, free on request. Send the batch number off the bag to kisan@van.com.pk or the helpline, and the certificate for that batch comes back. No sample to post.

3 · The laboratory

VAN Lab, PNAC LAB 336, accredited to ISO/IEC 17025:2017. Every batch is released through it before it ships. The certificate carries the laboratory's number.

What a certificate for VL-Potash Liquid shows

The declared analysis tested for the batch: K_2O against 30% w/v, with the batch number, the Pakistan Standard PS 5469-2020 and licence CM/L-2943/2023 on the bottle. Phosphorus and boron carry no label figure and are not certificate lines. No certificate for a VL-Potash Liquid batch was on file for this deck, so no batch reading is printed here; the grower asks for his own batch's certificate.

The lines a dealer says in Urdu.

The first line is the promise word from the VAN brand master. The other 3 are the deck's key sentences, translated in a first pass and not yet reviewed by VAN. Say them, and send corrections to partner@van.com.pk.

GUARD. The promise word for potash.

Urdu first pass, unreviewed

فصل کی ڈھال

When the crop starts to fill, give potash on the leaf.

Urdu first pass, unreviewed

جب فصل بھرنے لگے تو پتے پر پوٹاش دیں

This is the potash for the filling weeks. The season's potash is the bag in the ground.

Urdu first pass, unreviewed

یہ بھرائی کے دنوں کی پوٹاش ہے، موسم بھر کی پوٹاش زمین والا تھیلا ہے

Water first, then the bottle. Morning or evening.

Urdu first pass, unreviewed

پہلے ٹینک میں پانی، پھر بوتل۔ صبح یا شام

Write to the address for what you need.

If you grow

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch, a dealer near you.

If you sell

partner@van.com.pk

Dealers, distributors and the trade. Stock, training, corrections to this deck.

Everything else

info@van.com.pk

Supply, the laboratory, a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore. 1 helpline, 3 addresses. No group broadcasts: a grower who writes gets 1 reply.

VL-Potash Liquid: potash on the leaf or down the line, at the stage the crop fills. The basal bag is a different bag.

